

Hypogamy and Sorting

Frederik Almar¹ Yusuke Ishihata² Bastian Schulz¹ Pengpeng Xiao²

¹Aarhus University, ²Duke University

How to measure sorting? The contingency table

Male\Female	$t_{i,f} = 1$	$t_{i,f} = 2$	$\dots$	$t_{i,f} = N$	Marginal
$t_{i,m} = 1$	$P(1, 1)$	$P(1, 2)$	$\dots$	$P(1, N)$	$P(t_{i,m} = 1)$
$t_{i,m} = 2$	$P(2, 1)$	$P(2, 2)$	$\dots$	$P(2, N)$	$P(t_{i,m} = 2)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
$t_{i,m} = N$	$P(N, 1)$	$P(N, 2)$	$\dots$	$P(N, N)$	$P(t_{i,m} = N)$
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- 2 Women marrying up (men down) $\rightarrow$ Hypergamy (of women)
- 3 Women marrying down (men up) $\rightarrow$ **Hypogamy (of women)**

New ideas: a broader understanding of sorting patterns

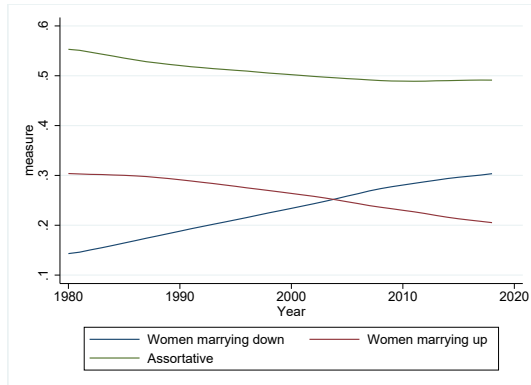
- The literature focuses on *same-type* couples (homophily).
- This takes into account only the main diagonal of the contingency table.
- Somewhat narrow. Same-type couples are typically a minority.
- Different-type couples are perhaps more important to understand, e.g., gender gaps and the link between sorting and inequality.

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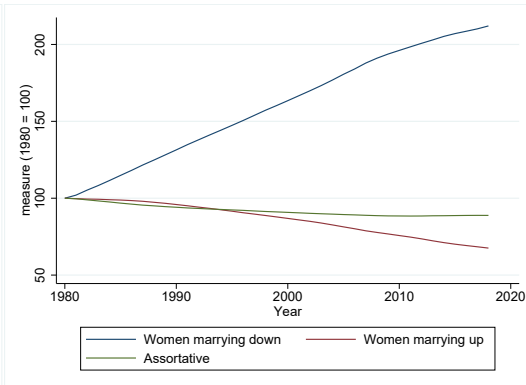
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- Different-type couples are perhaps more important to understand, e.g., gender gaps and the link between sorting and inequality.
- This project:
 - ① Develop new measures of hypergamy/hypogamy, study the trends.
 - ② Disentangle marriage and divorce margins. Is match formation or dynamic selection through the divorce margin relatively more important for the allocation of types?
 - ③ Explain the trends. Mechanical effects vs. preferences vs. changes in the labor market.

Trends in Marrying Up/Down and Homophily

Normalized trace



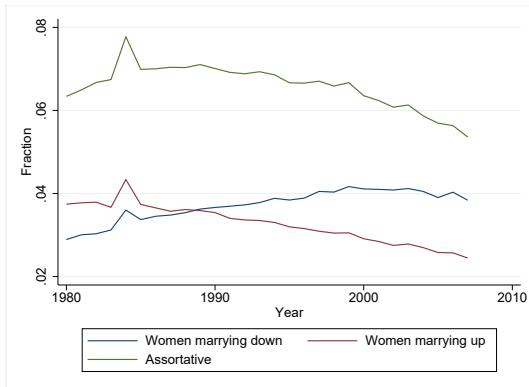
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Based on Likelihood Ratio

The Role of Marriage and Divorce

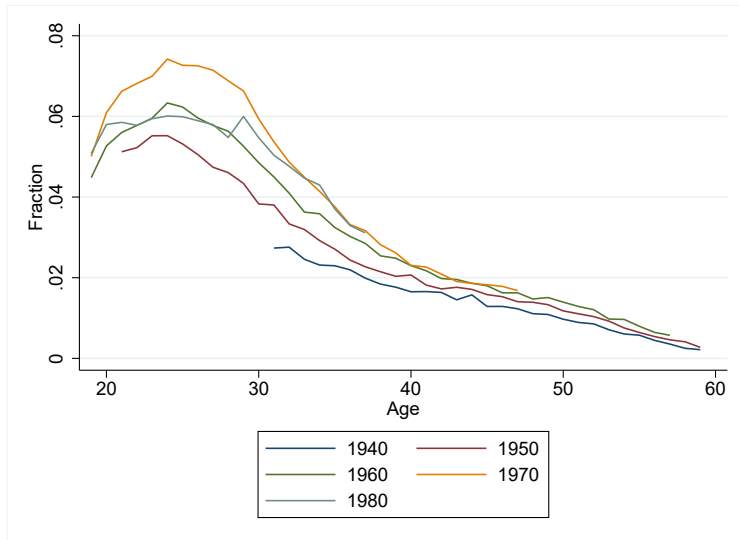
Marriage Rates (Inflow)



Divorce Rates (Outflow)

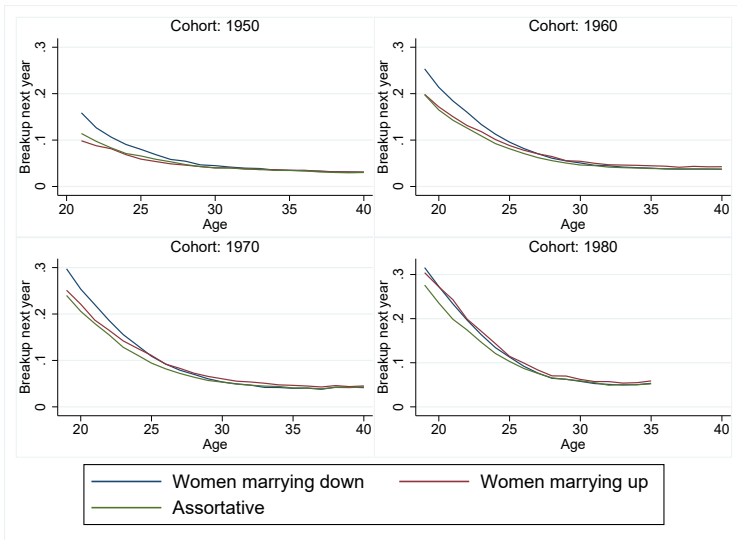


Inflow by Cohort



By cohort and education

Outflow by Cohort



Alternative

Outflow by Couple Type

Table: Divorce Rates by Education Level of Spouses

Male\Female	1980-2000				2001-2018				Change (%)			
	1	2	3	4	1	2	3	4	1	2	3	4
1	3.6	3.3	4.7	8.4	7.5	5.8	5.8	6.5	108	79	22	-22
2	2.8	2.2	3.2	6.1	5.9	4.0	3.6	4.2	112	84	14	-32
3	3.3	2.8	2.9	5.0	5.6	4.0	3.2	3.6	70	44	11	-27
4	3.8	3.3	3.0	3.5	5.0	3.9	3.1	2.8	32	16	3	-22

Let's write down a Structural Model

- Research question: why is there more hypogamy, i.e., why women are marrying down to a larger extent. Is this related to of gender convergence in the labor market?
- Not driven by the mechanical effects of changing marginal type distributions (LRs).
- Disentangle candidate explanations (non-exhaustive list):
 - ① Changing labor markets, e.g., narrower gender wage gaps.
 - ② Changing home production technologies, e.g., complementarities.
 - ③ Changing preferences, e.g., gender norms.
 - ④ Changing marriage and divorce “technologies”.
 - ⑤ First vs. later marriages.
- Ultimately, we want to understand the relative importance of these (and other) candidate explanations.

Model: Life-cycle (cohort) or Steady State (pooled cross sections)?

Some key questions that we are thinking about:

- How to model wages? Could be fixed types in a steady-state model (e.g., Goussé et al, 2017) or evolve endogenously over the life cycle (as in Almar et al., 2025).
- Changes in preferences might materialize across cohorts, so could be concealed in pooled cross sections. How to disentangle from changes in home production?
- We need endogenous marriage (standard) and divorce decisions (not so standard). Couple-specific divorce rates or evolving love shocks?

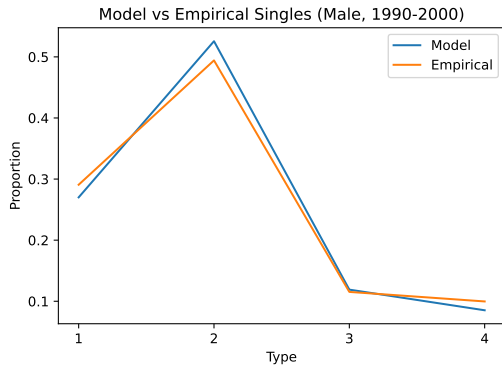
First pass: steady-state model with couple-specific divorce rates

Approach: combine MM version of Shimer & Smith (2000) with results in Goussé et al. (2017), Holzner & Schulz (2023), Foerster et al. (2025), and Borovickova & Shimer (2025)

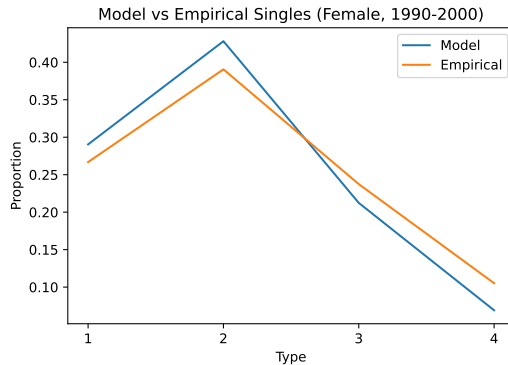
- Search-matching model of the marriage market, random search, four education-based types with fixed wages. Optimal time input choice of couples.
- Upon meeting, potential couples draw a love shock and only get married when it's high enough (selection margin).
- Updates of the shock trigger endogenous divorces at a rate that differs by couple type (due to differences in marital surplus).
- Today: separation shock, heterogeneous divorce rates differ by couple type, $\delta(x, y)$.

Model Fit: Marginal type distributions of Singles

Male

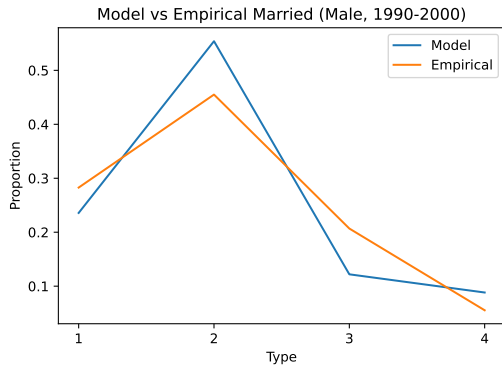


Female

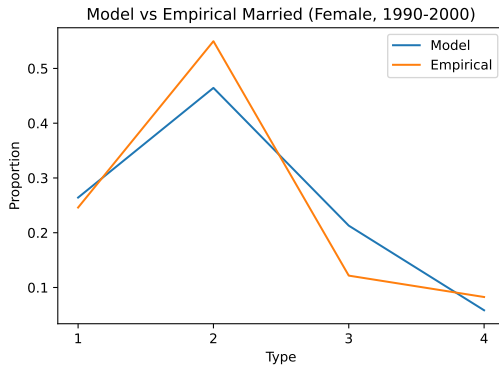


Model Fit: Marginal type distributions of Married

Male



Female



Conclusions and outlook

- Marriage market sorting patterns are changing. Not only for same-type couples.
- This matters for inequality across households and genders.
- Marginal type distributions change, but our measures take this into account.
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 - Increasing hypogamy of women might lead to more turnover in the marriage market.

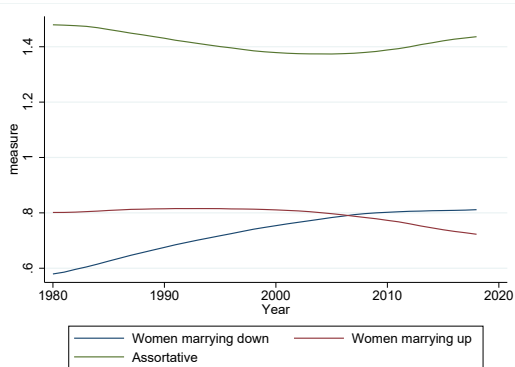
Thank you for your attention.

Bastian Schulz

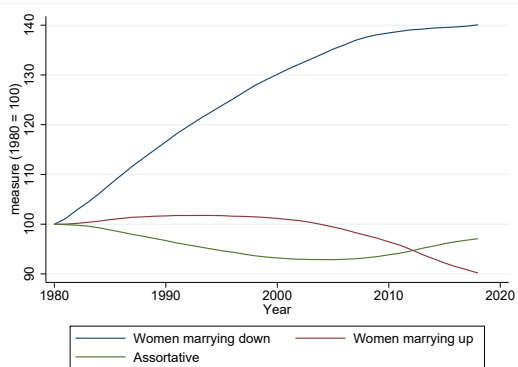
bastian.schulz@econ.au.dk

Likelihood Ratio

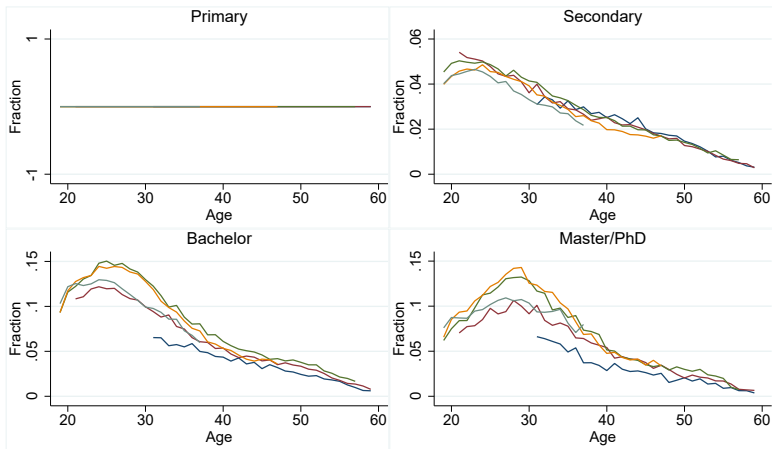
LR



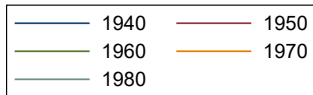
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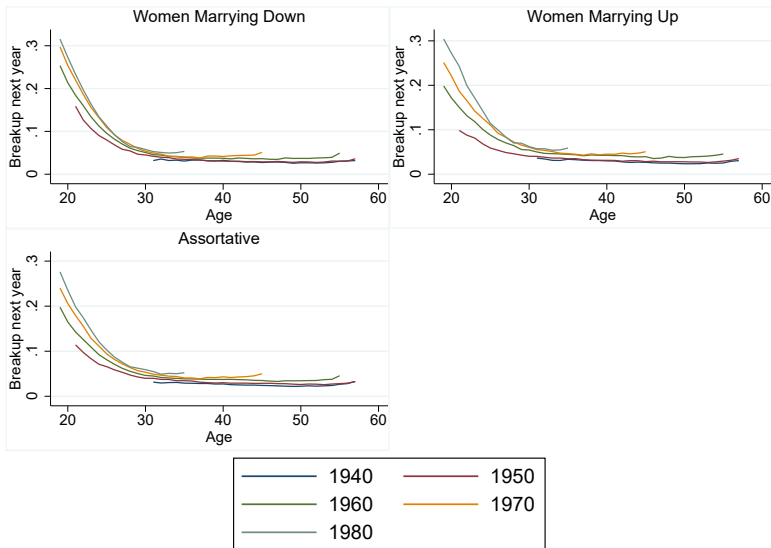
Inflow by Cohort and Education



► Go back



Outflow by Cohort (alternative)



► Go back